The documentation and process conversion measures necessary to comply with this document shall be completed by 30 April 2005.

INCH-POUND

MIL-PRF-19500/225G
30 December 2004
SUPERSEDING
MIL-PRF-19500/225F
25 June 1999

* PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, TYPES 2N1711, 2N1711S, 2N1890, AND 2N1890S, JAN AND JANTX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, low-power transistors. Two levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar to TO-5).

* 1.3 Maximum ratings unless otherwise specified T_A = +25°C.

Type (1)	P_T (2) T_C = +25°C	P_T-(2) T_A = +25°C	$R_{\theta JA}$ (3)	$R_{\theta JC}$ (3)	I_C	V_{CBO}	V_{CER} R_{BE} = 10Ω	V_{CEO}	V_{EBO}	T_J and T_{STG}
	W	W	°C/W	°C/W	mA dc	V dc	V dc	V dc	V dc	°C
2N1711	3.0	0.8	175	30	500	75	50	30	7	-65 to +200
2N1890	3.0	0.8	175	30	500	100	80	60	7	

(1) Also applies to the corresponding "S" suffix device.
(2) For derating see figures 2 and 3.
(3) For thermal impedance curves see figures 4 and 5.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

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1.4 Primary electrical characteristics.

Limits	h_{FE1} (1)/	h_{FE2} (1)	$\|h_{fe}\|$	$V_{CE(SAT)}$		
	V_{CE} = 10 V dc I_C = 10 μA dc	V_{CE} = 10 V dc I_C = 150 mA dc	f = 20 MHz V_{CE} = 10 V dc I_C = 50 mA dc	2N1711, S (1)	2N1890, S (1)	2N1890, S (1)
				I_C = 150 mA dc I_B = 15 mA dc	I_C = 150 mA dc I_B = 15 mA dc	I_C = 50 mA dc I_B = 5.0 mA dc
				V dc	V dc	V dc
Min	20	100	3.5	0.2	~~0.2~~	0.2
Max		300	12	1.5	5.0	1.2

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

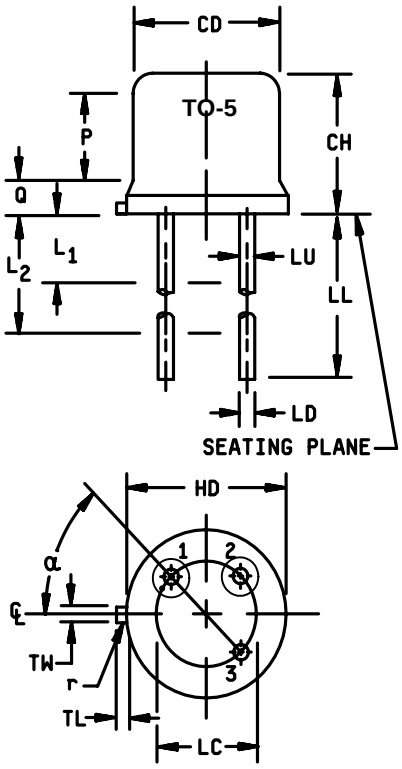
MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Ltr	Dimensions				
	Inches		Millimeters		Notes
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
LC	.200 TYP		5.08 TYP		7
LD	.016	.021	0.41	0.53	6
LL	See notes 7, 9, and 10				
LU	.016	.019	0.41	0.48	7
L1		.050		1.27	7
L2	.250		6.35		7
P	.100		2.54		5
Q		.050		1.27	
r		.010		0.254	8
TL	.029	.045	0.74	1.14	4
TW	.028	.034	0.71	0.86	3
α	45° TP		45° TP		6
Term 1	Emitter				
Term 2	Base				
Term 3	Collector				



NOTES:

1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Beyond r maximum, TW must be held to a minimum length of .021 inch (0.53 mm).
4. TL measured from maximum HD.
5. CD shall not vary more than ±.010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane .054 - .055 inch (1.37 - 1.40 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at a maximum material condition (MMC) relative to the tab at MMC. The device may be measured by direct methods or by gauge and gauging procedure.
7. LU applies between L1 and L2. LD applies between L2 and L minimum. Diameter is uncontrolled in L1 and beyond LL minimum.
8. r (radius) applies to both inside corners of tab.
9. For transistor types 2N1711S through 2N1890S, LL is .500 inch (12.70 mm) minimum, and .750 inch (19.50 mm) maximum (TO-5).
10. For transistor types 2N1711 through 2N1890, LL is 1.500 inches (38.10 mm) minimum, and 1.750 inches (44.45 mm) maximum (TO-5).
11. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions (TO-5).

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3. REQUIREMENTS

* 3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (TO-5).

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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* 4.3 Screening (list applicable JANTX level). Screening shall be in accordance with table IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement JANTX level
(1) 3c	Thermal impedance (see 4.3.2), method 3131 of MIL-STD-750.
7	Optional
9	Not applicable
11	h_{FE2}; I_{CBO1}
12	See 4.3.1
13	ΔI_{CBO1} = 100 percent of initial value or 5 nA dc; whichever is greater; Δh_{FE2} = ±15 percent of initial value;
14	Required

(1) Thermal impedance shall be performed anytime after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.

* 4.3.1 Burn-in conditions. Burn-in conditions are as follows: V_{CB} = 10 - 30 V dc. Power shall be applied to achieve T_J = +135°C minimum using a minimum P_D = 75 percent of P_T maximum, T_A ambient rated as defined in 1.3.
NOTE: No heat sink or forced air cooling of the devices shall be permitted.

* 4.3.2 Thermal impedance (measurements). The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} (and V_C where appropriate). The thermal impedance limit shall comply with the thermal impedance graph in figures 4 and 5 (less than or equal to the curve value at the same t_H time) and shall be less than the process determined statistical maximum limit as outlined in method 3131. See group E subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. The test conditions for $Z_{\theta JX}$ shall be those used in 4.3.2.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified in paragraph 4.4.2.1 herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, JAN and JANTX.

Step	Method	Condition
1	1026	Steady-state life: Test condition B, 1,000 hours minimum, V_{CB} = 10 V dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased so long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is a minimum of 340 hours.
2	1048	Blocking life, T_A = +150°C, V_{CB} = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

* 4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN and JANTX samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot.

b. Must be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (group B for JAN and JANTX) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Lead fatigue: Test condition E.
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3 and 4.3.2).
C6	1026	T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5). T_J = 150°C min; V_{CB} = 24 V dc for 2N1711 and 2N1711S; V_{CB} = 48 V dc for 2N1890 and 2N1890S.

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* 4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests herein for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

* 4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

* 4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical examination 3/	2071	n = 45 devices, c = 0				
Solderability 3/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4//	1022	n = 15 devices, c = 0				
Temp cycling 3/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Electrical measurements		Table I, subgroup 2				
Hermetic seal 5/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Bond strength 3/	2037	Precondition T_A = +250°C at t = 24 hours or T_A = +300°C at t = 2 hours n = 11 wires, c = 0				
Decap internal visual (design verification)	2075	n = 4 devices, c = 0				
Subgroup 2						
Thermal impedance	3131	See 4.3.2	$Z_{\theta JX}$			°C/W
Breakdown voltage, collector to base	3001	Bias condition D; I_C = 100 μA dc;	$V_{(BR)CBO}$			
2N1711, 2N1711S				75		V dc
2N1890, 2N1890S				100		V dc
Breakdown voltage, emittor to base	3026	Bias condition D; I_E = 100 μA dc;	$V_{(BR)EBO}$	7		V dc
Breakdown voltage, collector to emiiter	3011	Bias condition D; I_C = 30 mA dc; Pulsed (see 4.5.1).	$V_{(BR)CEO}$			
2N1711, 2N1711S				30		V dc
2N1890, 2N1890S				60		V dc
Breakdown voltage, collector to emiiter	3011	Bias condition D; I_C = 100 mA dc; Pulsed (see 4.5.1). R_{BE} = 10Ω.	$V_{(BR)CER}$			
2N1711, 2N1711S				50		V dc
2N1890, 2N1890S				80		V dc

See footnotes at end of table.

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* TABLE I. Group A inspection. Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 – Continued.						
Collector to base cutoff current	3036	Bias condition D	I_{CBO1}			
2N1711, 2N1711S		V_{CB} = 60V dc			10	nA dc
2N1890, 2N1890S		V_{CB} = 80 V dc			10	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO}		5.0	nA dc
Collector-emitter saturation voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$			
2N1711, 2N1711S					1.5	V dc
2N1890, 2N1890S					5.0	V dc
Collector-emitter saturation voltage	3071	I_C = 50 mA dc; I_B = 5 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$			V dc
2N1890, 2N1890S					1.2	V dc
Base emitter saturation voltage	3066	Test condition A, I_C = 150 mA dc, I_B = 15 mA dc, pulsed (see 4.5.1).	$V_{BE(sat)1}$		1.3	V dc
Base emitter saturation voltage	3066	Test condition A, I_C = 50 mA dc, I_B = 5 mA dc, pulsed (see 4.5.1).	$V_{BE(sat)2}$			V dc
2N1890, 2N1890S					0.9	V dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 µA dc pulsed (see 4.5.1).	h_{FE1}	20		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc pulsed (see 4.5.1).	h_{FE2}	100	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc pulsed (see 4.5.1).	h_{FE3}			
2N1711, 2N1711S				50		
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D;	I_{CBO2}			
2N1711, 2N1711S		V_{CB} = 60 V dc			10	µA dc
2N1890, 2N1890S		V_{CB} = 80 V dc			15	µA dc
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc pulsed (see 4.5.1).	h_{FE4}	35		

See footnotes at end of table.

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* TABLE I. Group A inspection. Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Small-signal short-circuit forward current transfer ratio	3206	V_{CE} = 5 V dc; I_C = 1 mA dc f = 1KHz	h_{fe}	80	200	
		V_{CE} = 10 V dc; I_C = 5 mA dc f = 1KHz.	h_{fe}	90	270	
Magnitude of small-signal short-circuit forward current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 50 mA dc; f = 20 MHz	$\lvert h_{fe} \rvert$	3.5	12	
Small-signal short- circuit input impedance	3201	V_{CB} = 10 V dc; I_C = 5 mA dc	h_{ib}	4	8	ohms
Small-signal short-circuit input Admittance	3216	V_{CB} = 10 V dc; I_C = 5 mA dc	h_{ob}			
2N1711, 2N1711S 2N1890, 2N1890S				0.0 0.0	1.0 .3	µmhos µmhos
Small-signal open-circuit reverse voltage transfer ratio	3211	V_{CB} = 10 V dc; I_C = 5 mA dc	h_{rb}			
2N1711, 2N1711S 2N1890, 2N1890S					$5x10^{-4}$ $1.5x10^{-4}$	µmhos µmhos
Open circuit output capacitance		V_{CB} = 10 V dc; I_E = 0 mA dc 100kHz ≤ f ≤ 1 mHz	C_{obo}			
2N1711, 2N1711S 2N1890, 2N1890S				8 5	25 15	pF pF
Pulse response	3251	Test condition A, except test circuit and pulse requirements. See figure 6 herein.	$t_{on} + t_{off}$		30	ns
Subgroup 5, 6, and 7						
Not applicable						

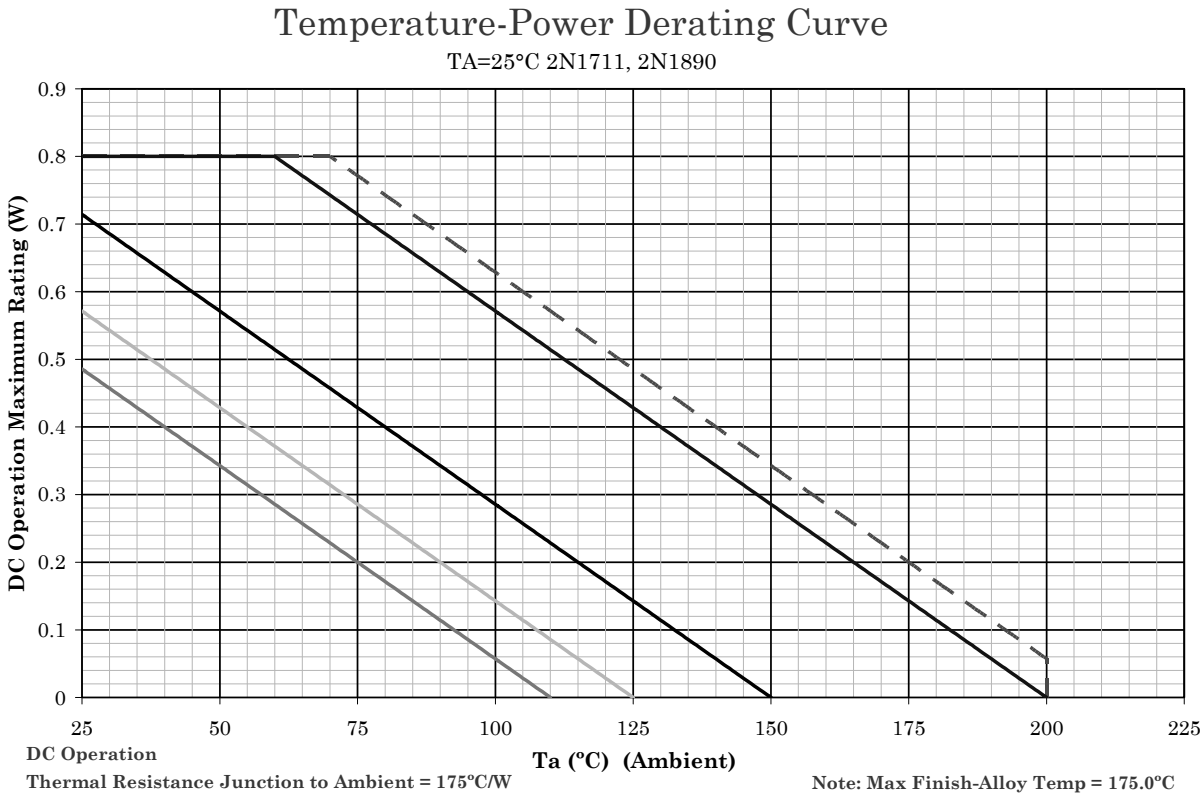
1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed test in subgroup 1 of table I, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for laser marked devices.
5/ This hermetic seal test is an end-point to temp-cycling in addition to electrical measurements.

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* TABLE II. Group E inspection (all quality levels) - for qualification or re-qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2 and 4.5.2 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = 10 V dc, 6,000 cycles	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 4			
Thermal impedance curves		Each supplier shall submit their qualification lot average design maximum thermal impedance curves. In addition, the optimal test conditions and thermal impedance limit shall be provided to the qualifying activity in the qualification report.	Sample size N/A
Subgroup 5			
Not applicable			
Subgroup 6			3 devices
Electrostatic discharge (ESD)	1020		
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition A for devices ≥ 400 V Condition B for devices < 400 V.	

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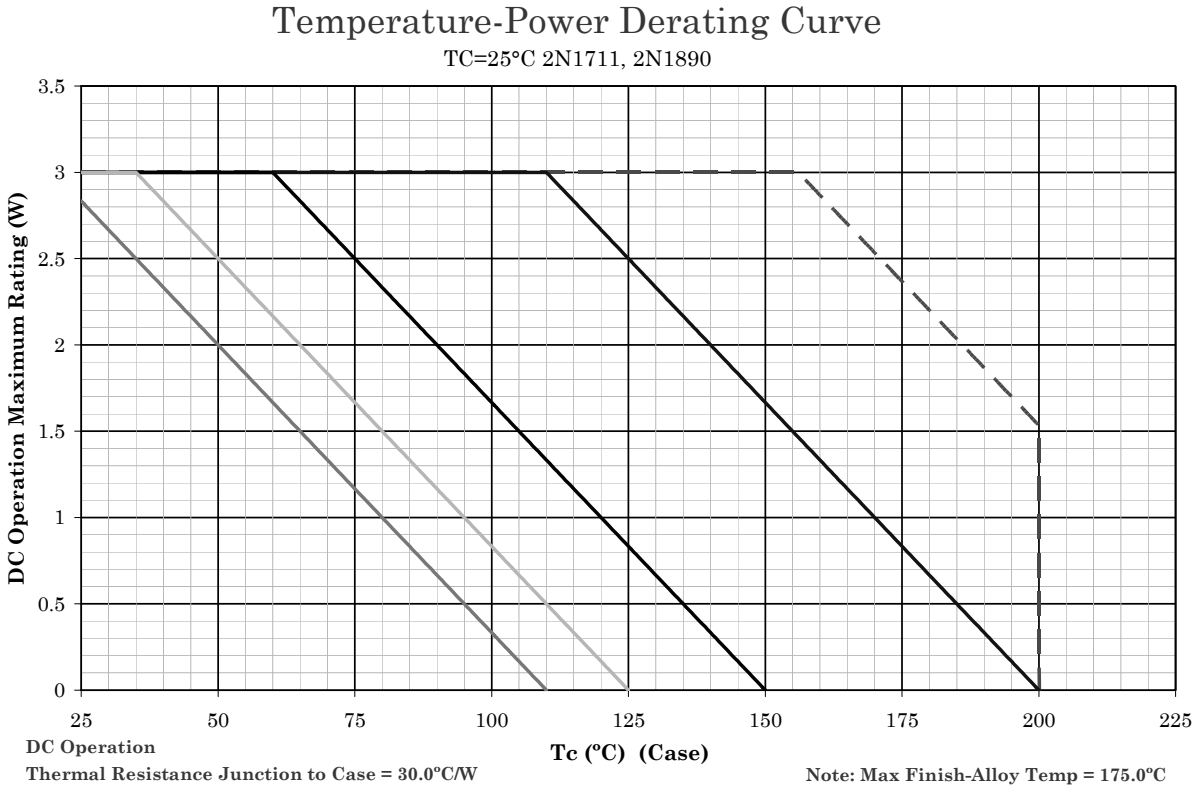


NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

* FIGURE 2. Temperature-power derating for 2N1711, 2N1711S, 2N1890, and 2N1890S (TO-5).

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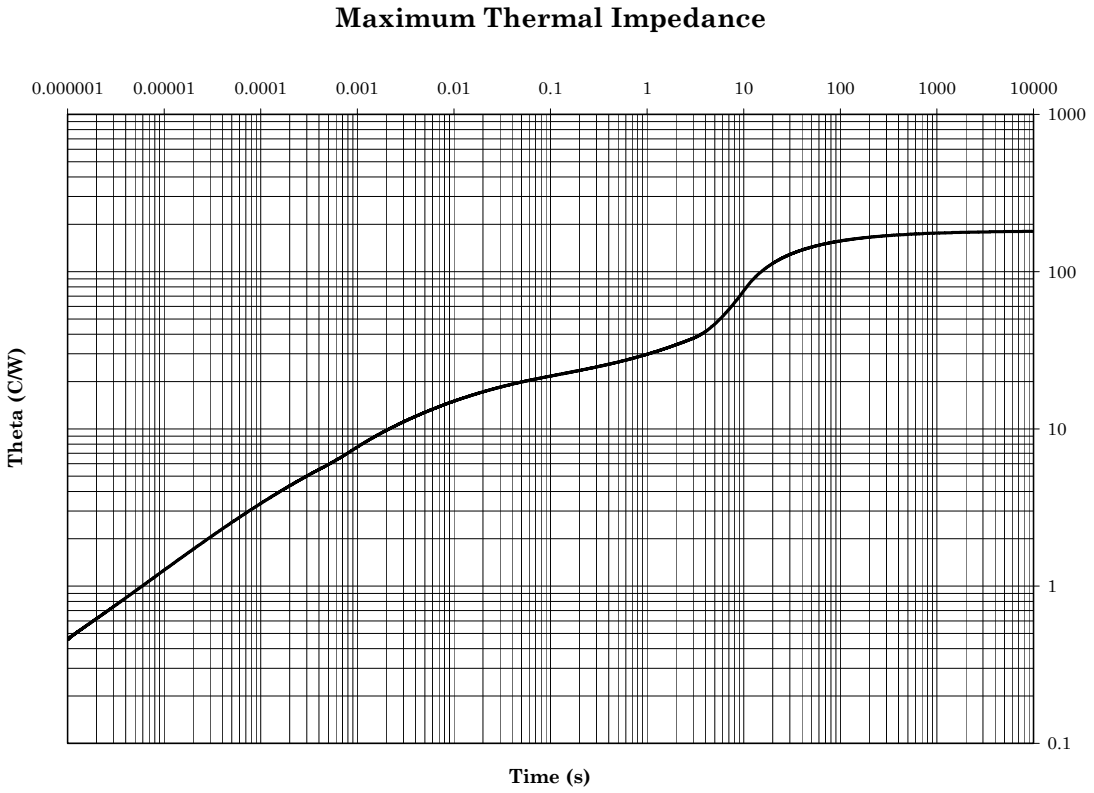


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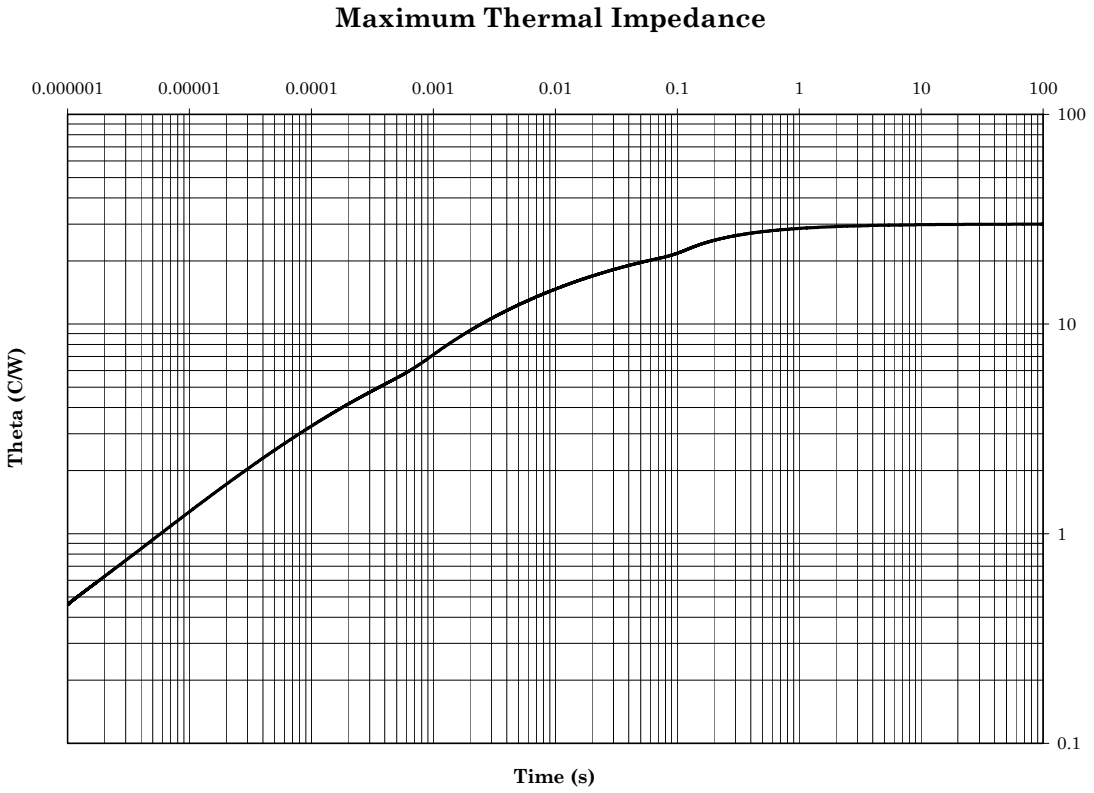
* FIGURE 3. Temperature-power derating for 2N1711, 2N1711S, 2N1890, and 2N1890S (TO-5).

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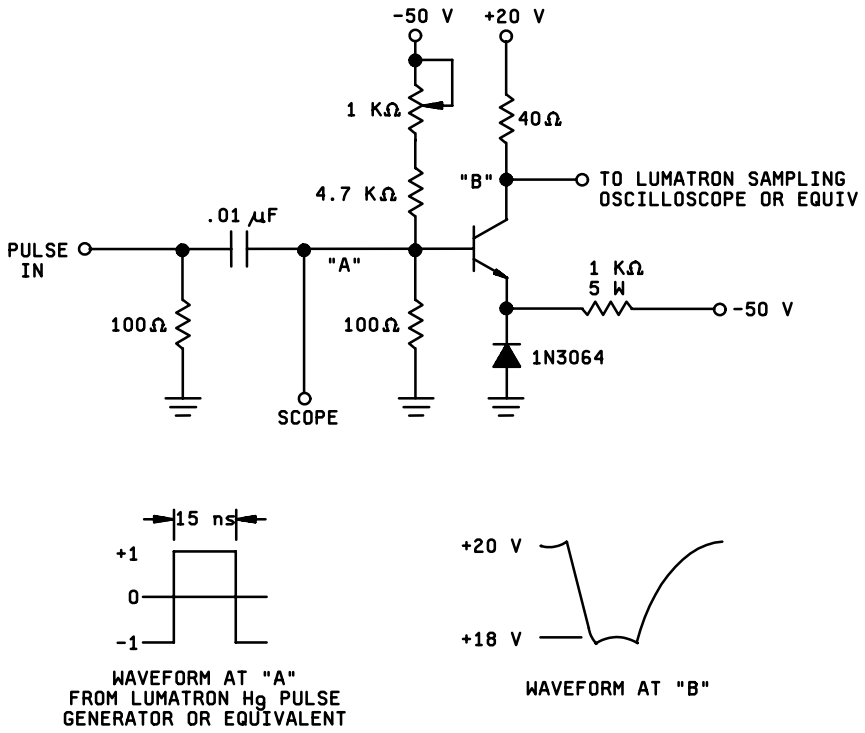
* FIGURE 4. Thermal impedance graph ($R_{\theta JA}$) for 2N1711, 2N1711S, 2N1890, and 2N1890S (TO-5).

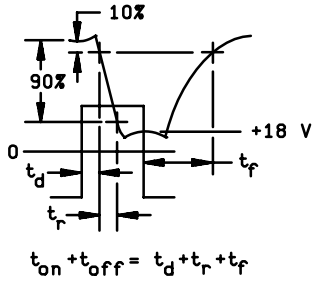
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* FIGURE 5. Thermal impedance graph ($R_{\theta JC}$) for 2N1711, 2N1711S, 2N1890, and 2N1890S (TO-5).

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NOTES:
1. The rise time (T_r) of the applied pulse shall be 2.0 ≤ ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50 Ω.
2. Sampling oscilloscope: $Z_{in} \geq 100$ kΩ, $C_{in} \leq 12$ pF, rise time ≤ 2 ns.

FIGURE 6. Pulse response (turn-on plus turn-off) measurement circuit and waveforms.

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
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Air Force - 11
NASA - NA
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Review activities:
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Preparing activity:
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(Project 5961-2956)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/ .

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